

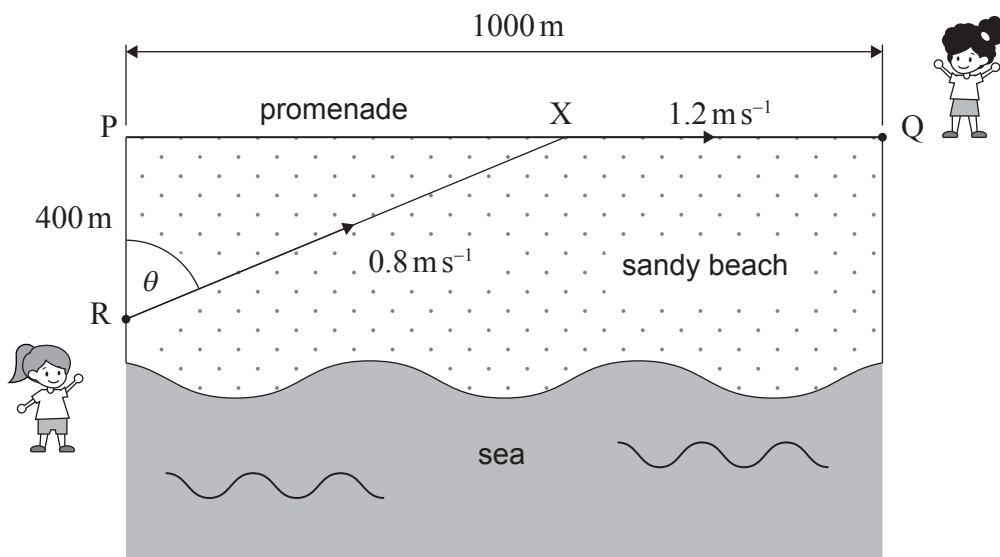
8. [Maximum mark: 9]

Astrid and Bronwyn are on vacation at Blackpool beach.

Astrid is standing on the beach at point R and sees Bronwyn standing on the promenade at point Q. $PR = 400\text{ m}$, $PQ = 1000\text{ m}$, $\angle RPX = \theta$ where $0 < \tan \theta \leq \frac{5}{2}$.

Astrid walks in a straight line from R with speed 0.8 m s^{-1} until reaching point X on the promenade. She then jogs along the promenade with speed 1.2 m s^{-1} .

This is shown in the following diagram.



It is given that $RX = 400 \sec \theta$ and $XQ = 1000 - 400 \tan \theta$.

The time, in seconds, for Astrid to reach Bronwyn is given by T .

(a) Show that $T = 500 \sec \theta + \frac{2500 - 1000 \tan \theta}{3}$. [2]

(b) Find $\frac{dT}{d\theta}$. [3]

Astrid chooses the angle θ of her walk across the beach to point X, in order to reach Bronwyn in the shortest time possible. You may assume that T has exactly one minimum value.

(c) Show that in this case $PX = 160\sqrt{5}\text{ m}$. [4]

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(Question 8 continued)

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